QUALITY ASSURANCE TESTING DOCUMENTATION

(QATD)

UMHackathon 2026

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Document Control

Field	Detail
System Under Test (SUT)	UMHackathon 2026 - Team **Ctrl + She**
Team Repo URL	https://github.com/dhvstr/Hackathon-2026
Project Board URL	https://trello.com/invite/b/69ece804f15c863d1f5181ba/ATTI21bf5720a2b64be93be5ee64d74ce5359D30C3CD/smezy-um-hackathon-2026

Objective:

The main objective of this system is to make sure the AI SME Financial Decision Intelligence System can work smoothly and reliably. It should be able to handle financial data, run simulations, give AI-based explanations, and provide clear recommendations to users. The system must also stay stable when many users are using it at the same time and when it connects with the AI service (Gemini and Z.AI GLM).

PRELIMINARY ROUND (Test Strategy & Planning)

1. Scope & Requirements Traceability

This section makes sure all testing is linked back to the system requirements. It helps ensure every main feature is tested properly so nothing important is missed and no extra unwanted features are added.

1.1 In-Scope Core Features

- **Financial Data Input**
  Users can enter and manage financial data like cash, expenses, and invoices.

- **Cash Flow Simulation**
  System calculates and shows the impact of user decisions on cash flow.

- **AI Insight and Recommendation**
  System uses AI to explain results and give financial suggestions.

- **Document Analysis**
  Users can upload financial documents, and the system extracts key information and insights.

1.2 Out-of-Scope

- **Profile Settings**
  Users cannot edit profile details like name, picture, or personal preferences.

- **App Theme / UI Customization**
  Users cannot change the app theme, colors, or layout design.

- **Social Sharing Features**
  The system does not support sharing results or insights to social media or external platforms.

- **Marketing and Promotion Tools**
  Features like ads, promotions, or marketing campaigns are not included.

- **Real-Time Banking Integration**
  The system does not connect directly to real bank accounts or live banking APIs.

2. Risk Assessment & Mitigation Strategy

Technical Risk	Likelihood (1–5)	Severity (1–5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach
AI generates incorrect financial recommendation	3	5	15 (High)	Add validation layer to check AI output consistency and restrict unsafe recommendations	Test with misleading inputs and check AI response accuracy
Simulation gives wrong cash flow result	2	5	10 (Medium)	Use fixed calculation rules and double-check formulas in backend logic	Compare manual calculation vs system output
AI service API timeout or failure	3	4	12 (High)	Implement retry mechanism and fallback response message	Simulate API downtime and observe system behavior
Incorrect or incomplete document extraction	3	4	12 (High)	Improve prompt design and apply structured output format from AI	Upload different PDF formats and check extracted result
Data loss during simulation process	2	5	10 (Medium)	Use database transaction handling and auto-save state	Force system crash during simulation and check recovery
Slow system response under high user traffic	4	4	16 (Critical)	Optimize backend processing and use caching for frequent requests	Load testing with multiple simultaneous users

Risk Assessment Scoring Criteria:

Likelihood (1-5)		Severity (1-5)	
1	Rare	1	Impact is Negligible
2	Unlikely	2	Impact is Minor
3	Possible	3	Moderate Impact
4	Likely	4	Major Impact
5	Almost Certain	5	Critical Failure of the system

Risk Score = Likelihood × Severity

Risk Level Reference:

Risk Score	**Risk Level**	**Recommended Action**
1 – 5	Low	Monitor only. Acceptable risks.
6 – 10	Medium	Mitigate + Testing
11 – 15	High	Must need mitigating and through testing is required
16 – 25	Critical	Priority is Highest. Need extensive level of testing.

3. Test Environment & Execution Strategy

This section explains where the system will be tested, how test data is handled, and how testing is controlled during different phases of development. It ensures the system is properly validated before going live.

- **Unit Test**
  - Testing focuses on individual components such as financial data validation, cash flow simulation logic, AI prompt generation, AI response parsing, and document insight extraction.
  - Tests are written using PyTest and run during development. They are also automatically executed in the CI pipeline whenever code is pushed to the repository.
  - Each component is tested separately. External dependencies such as AI service layer API and database are mocked to ensure only the core logic is tested.
  - Tests pass when the system correctly handles valid inputs, rejects invalid inputs, and produces expected outputs. For example, invalid cash input should return “Invalid amount format”, and simulation output should match expected cash flow results.

- **Integration Test**
  - Testing focuses on interaction between major components such as Frontend, Simulation Service, AI Insight Service, and Database.
  - Performed after unit testing is completed. It involves merging services and testing real API communication between modules.
  - Real API calls are used (no mocking). The system is tested to ensure data flows correctly from financial input to simulation engine to AI service to final output.
  - Test is passed when all connected services work together correctly and the system returns accurate simulation results and AI recommendations.

- **Test Environment (CI/CD Practice):**
  - **Local Testing:**
    Developers manually test the system on localhost during development.
  - **Staging / CI Environment:**
    GitHub Actions is used to automatically run tests whenever code is pushed or a pull request is created.
  - **CI/CD Pipeline:**
    Automated pipeline runs unit tests and integration tests. Only code that passes all checks is allowed to merge into the main branch.

- **Regression Testing & Pass/Fail Rules:**
  - **Execution Phase**
    Regression testing is done every time new features or updates are added. It ensures that existing features like simulation, AI insights, and document analysis still work correctly.
  - **Pass/Fail Condition**
    A test is considered passed only when the actual output matches the expected output exactly. Any mismatch is marked as failed and recorded in the defect log for fixing.
  - **Continuation Rule**
    Testing can only proceed to the next phase if all critical test cases pass the required threshold. If not, development must fix issues before continuing.

- **Test Data Strategy:**
  - **Manual:**
    Mock SME financial profiles are created, such as businesses with low cash flow, high expenses, loan dependency, and unpaid invoices.
  - **Automated:**
    SQL seed scripts are used to generate dummy financial data such as cash records, expenses, invoices, and simulation scenarios for testing purposes.

- **Passing Rate Threshold:**
  - At least 85% of all test cases must pass for the system to be accepted.
  - For critical components such as cash flow simulation, AI recommendation, and financial decision output, the pass rate must be 100% with no exceptions, as failures in these areas directly affect business decisions.

4. CI/CD Release Thresholds & Automation Gates

This section is mainly for defining metrics & threshold for success criteria. So, if a certain threshold is not met, the pipeline blocks the forward transition.

4.1 Integration Thresholds (Merging to Main)

The primary needs of this threshold to be crucial is because the main branch must be a stable source of truth. So, it needs to be achieved through continuous integration checks.

Checks	Requirements	Project	Pass/Failed
Automatic Build	Zero Build Error	0 Errors	Passed
Unit Tests	100% Passing Rate	100%	Passed
Code Quality	Zero Linting Error	95%	Failed
Test Coverage	Minimum 80%	78%	Failed

4.2 Deployment Thresholds (Pushing to Production)

Checks	Requirements	Project	Pass/Failed
Regression Test	Minimum 90%	92%	Passed
AI Output Pass Rate	Minimum of 80% of documented prompt	79%	Failed
Critical Bugs	Zero P0/01 Bugs	1	Failed
API Performance	Response Time < 800ms	420ms	Passed
Security	API keys are not exposed	Clean	Passed

5. Test Case Specifications (Drafts)

This section defines the main test cases used to validate the system. It includes one full user journey (happy case), one error/edge case, and two performance-based non-functional tests.

Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
TC-01	**Happy Case (Entire Flow):** Cash Flow Simulation + AI Insight	Verify that a user can input financial data, run a chatbot simulation, and receive AI-generated recommendation successfully.	1. User logs into the system 2. Enters financial data (cash, expenses, invoices) 3. Selects scenario 4. Clicks "Run Simulation" 5. System processes cash flow impact 6. AI generates explanation and recommendation 7. Result is displayed to user	System should show correct cash flow change, risk level, and AI recommendation clearly. All components (simulation + AI + UI) must work together smoothly.	Simulation runs successfully. Cash flow updated correctly. AI provides recommendation: "Delay supplier payment by 14 days to maintain liquidity." Status: Passed
TC-02	**Specific Case (Negative):** Financial Data Validation	Verify system handles invalid input such as negative cash value or empty fields without crashing.	1. Open financial input form 2. Enter negative cash value (-5000) 3. Leave expense field empty 4. Click "Run Simulation"	System should block submission and show error message like "Invalid financial input. Please enter valid values."	System blocked input but did not show clear error message. No crash occurred, but user feedback is unclear. Status : Failed
TC-03	**NFR (Performance):** Simulation + AI Response	Measure system response time when multiple users run financial simulations at the same time.	1. Use Postman / load testing tool 2. Simulate 50 concurrent users 3. Each user triggers "Run Simulation" API 4. Measure average response time	System response time should be under 800ms with no errors.	Average response time recorded at 1050ms under load. No system crash, but slower than expected Status : Failed

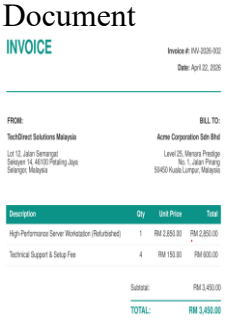
TC-04	**NFR (AI Output Quality):** AI service layer Recommendation	Check whether AI generates accurate and relevant financial recommendations based on simulation results.	1. Run multiple simulation scenarios 2. Send results to AI service API 3. Compare AI output with expected financial logic	At least 80% of AI recommendations must be correct, relevant, and aligned with financial outcomes	78% of outputs are accurate. Some recommendation are too generic or partially incorrect. Status : Failed

6. AI Output & Boundary Testing (Drafts)

6.1. **Prompt/Response Test Pairs**

Test ID	**Prompt Input**	**Expected Output (Acceptance Criteria)**	**Actual Output**	**Status**
AI-01 (AI Insight Generator)	Ringgit nears 4-year high as economy gains strength	The output should have action recommendations of less than 40 words and clear explanations of less than 40 words.	Recommendation: Consider importing premium ingredients or kitchen equipment now. The stronger Ringgit lowers your purchasing costs for imported goods, allowing for better margins on high-end catering packages. (28 words) Explanation: A stronger Ringgit means your local currency has more buying power against foreign currencies. This reduces the cost of imported supplies, directly improving your bottom line and cash reserves. (31 words)	Pass
AI-02 (AI Insight Generator)	SST increases	The output should have action recommendations of less than 40 words and clear	Recommendation: Review your menu pricing immediately. Update your service	Pass

		explanations of less than 40 words.	contracts to reflect the new tax rate and inform regular clients of price adjustments to maintain your current profit margins. (29 words) Explanation: Sales and Service Tax (SST) increases raise your operational costs and the final price for customers. Failing to adjust prices will result in the business absorbing the tax, reducing cash flow. (33 words)	
AI-03 (AI Insight Generator)	Govt weighs diesel Road Tax cut to ease fuel cost pressure	The output should have action recommendations of less than 40 words and clear explanations of less than 40 words.	Recommendation: Maintain your existing delivery fleet instead of downsizing. Use the anticipated tax savings to offset rising raw material costs or reinvest in more efficient delivery scheduling software. (28 words) Explanation: A reduction in diesel road tax lowers the fixed overhead costs for your delivery vans. This provides	Pass

			immediate relief to your logistics budget, improving monthly net cash flow. (31 words)	
AI-04 (Document Scanner)	Invoice Pdf Document  INVOICE Invoice #: INV-2026-002 Date: April 22, 2026 FROM: TechDirect Solutions Malaysia Lot 12, Jalan Semangat Seksyen 14, 46100 Petaling Jaya Selangor, Malaysia BILL TO: Acme Corporation Sdn Bhd Level 25, Menara Prestige No. 1, Jalan Pinang 50450 Kuala Lumpur, Malaysia Description Qty Unit Price Total High-Performance Server Workstation (Refurbished) 1 RM 2,850.00 RM 2,850.00 Technical Support & Setup Fee 4 RM 150.00 RM 600.00 Subtotal: RM 3,450.00 TOTAL: RM 3,450.00	The output should extract data such as Vendor, Invoice Date and Amount. It also give recommendation on the next action (Pay now, Negotiable invoice)	Vendor: TechDirect Solutions Malaysia Invoice Date: April 22, 2026 Amount: 3450.0	Pass
AI-05 (Document Scanner)	Invoice Pdf Document INVOICE Invoice #: INV-2026-001 Date: April 20, 2026 FROM: Lumina Strategic Consulting Group Lot 12, Jalan Semangat Seksyen 14, 46100 Petaling Jaya Selangor, Malaysia BILL TO: Acme Corporation Sdn Bhd Level 25, Menara Prestige No. 1, Jalan Pinang 50450 Kuala Lumpur, Malaysia Description Qty Unit Price Total Strategic Business Transformation Consulting - Phase 1 1 RM 20,000.00 RM 20,000.00 Market Analysis & Decision Intelligence Implementation 1 RM 10,000.00 RM 10,000.00 Subtotal: RM 30,000.00 TOTAL: RM 30,000.00	The output should extract data such as Vendor, Invoice Date and Amount. It also give recommendation on the next action (Pay now, Negotiable invoice)	Vendor: Lumina Strategic Consulting Group Invoice Date: April 20, 2026 Amount: 30000.0	Pass
AI-06 (Document Scanner)	Invoice Pdf Document SME SOLUTIONS LTD. INVOICE 123 Business Avenue, Shah Alam Selangor, Malaysia contact@smesolutions.com Invoice #: INV-2026-0042 Date: April 24, 2026 Due Date: May 08, 2026 BILL TO Global Tech Partners 45 Innovation Drive Cyberjaya, Malaysia Attn: Accounts Payable PROJECT REFERENCE Q3 Digital Transformation Phase 1 Implementation PO #: 88921-07 Description Quantity Unit Price (RM) Total (RM) Business Intelligence Dashboard Development 1 5,500.00 5,500.00 Data Integration & Cleaning (Structured/Unstructured) 15 120.00 1,800.00 ZAI GUM Critical Path Integration 1 2,200.00 2,200.00 System Analysis Documentation & Staff Training 1 1,000.00 1,000.00 Subtotal: RM 10,500.00 SST (6%): RM 630.00 TOTAL AMOUNT DUE: RM 11,130.00	The output should extract data such as Vendor, Invoice Date and Amount. It also give recommendation on the next action (Pay now, Negotiable invoice)	Vendor: Global Tech Partners Invoice Date: May 08, 2026 Amount: 11130.0	Pass

6.2. Oversized/Larger Input Test

Fields	Details
Maximum Input Size	1500 tokens
Input used while testing	6000 words, 8000 tokens
Expected Behavior	The system should reject the input with a user-friendly error message or truncate the text to the maximum limit before processing to prevent API timeout and high latency.
Actual Behavior	The API Gateway identified the payload exceeded the max_token constraint. The system blocked the request and returned a “413 Payload Too Large” status with the message: "Input too long. Please upload a document under 3 pages or 1,500 tokens."
Status	Pass

6.3. Adversarial/Edge Prompt Test

Test ID	**Prompt Input**	**Expected Output (Acceptance Criteria)**	**Actual Output**	**Status**
AI-01 (AI Insight Generator)	Why?	The output should have action recommendations of less than 40 words and clear explanations of less than 40 words.	You have any problem regarding SME business?	Pass
AI-02 (Document Scanner)	Non-Invoice Pdf Document	[e.g. All the JSON fields does reflect accurately and there is no hallucinated values]	Vendor: Not Fount Invoice Date: Not Found Amount: Not Found	Pass

6.4. Hallucinating Handling

The system reduces hallucination (false or made-up AI output) using several methods:

- The prompt always includes structured financial data from the simulation engine
- AI is not allowed to generate numbers freely; it must follow system-calculated values
- Output is validated before being shown to the user
- If response is unclear or inconsistent, system triggers a fallback message like: "Insufficient data to generate accurate recommendation"

Important Note:
All AI tests are done continuously during development using Agile practice to ensure the system improves step by step instead of only being tested at the end.